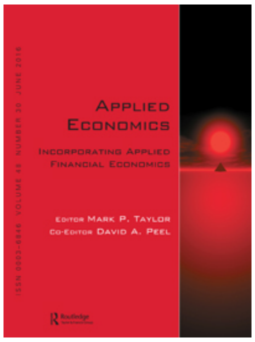

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# Forward-premium puzzle: is it time to abandon the usual regression?

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**ABSTRACT**

The forward premium puzzle is usually evidenced by the rejection of the null hypothesis in the uncovered interest parity (UIP) regression. Because this parity need only hold in a risk-neutral world, a risk adjustment term is missing from the equation if speculation in foreign exchange markets is risky. We deal with this issue following the literature which assumes that discounted returns on foreign government bonds are log-normal, so we can linearize the Euler pricing equations (in level) and obtain a modified UIP system for which the risk adjustment term is obtained by applying to the pricing kernel-based relations a generalized autoregressive conditional heteroscedasticity-in-mean model. However, here we innovate by adopting a methodology which differs from all these related works. We construct and use a stochastic discount factor that does not depend on a specific model, by residing in the space of returns which we extract from the data by simply imposing the orthogonality restrictions represented by the Euler equations. So, we devise a purely statistical pricing kernel that performs well in in-sample level equations. Somewhat disappointingly, the risk premium inclusion in the conventional regression changes neither the significance nor the magnitude of the forecasting power of the forward premium for most currencies we study. The contrasting performance of the tests in level and in logs suggests that linearization may be to blame.

**KEYWORDS**
Forward premium puzzle; return-based pricing kernel; bivariate GARCH-M; conditional log-normality

**JEL CLASSIFICATION**
G12; G15

## I. Introduction

Recall the uncovered interest parity (UIP) regression,

$$s_{t+1} - s_t = \alpha_0 + \alpha_1 ({}_t f_{t+1} - s_t) + u_{t+1}, \qquad (1)$$

where $s_t$ is the log of the exchange rate at time $t$, ${}_t f_{t+1}$ is the log of time $t$ forward exchange rate contract for delivery at time $t+1$ and $u_{t+1}$ is the regression error.

Notwithstanding the possible effect of Jensen inequality terms, rational expectations and risk neutrality implies that $\alpha_1 = 1$ and $\alpha_0 = 0$, the linear expression of the UIP. Testing the UIP is, therefore, equivalent to testing the null $\alpha_1 = 1$ and $\alpha_0 = 0$, along with the lack of correlation of residuals from Equation (1).

Most studies report $\hat{\alpha}_1$ significantly smaller than *zero*. The forward premium, defined as the difference between the logarithm of the forward rate, ${}_t f_{t+1}$, and the logarithm of the current exchange rate, $s_t$, is too strongly (negatively) correlated with subsequent changes in the (log of the) exchange rate, $s_{t+1} - s_t$, in the data. As a result, domestic currency is oddly expected to appreciate when domestic nominal interest rates exceed foreign interest rates. Interest rates differentials 'predict' differential returns. This is the forward premium puzzle (FPP) in a nutshell.[1]

Note that the uncovered parity need only hold exactly in a world of risk-neutral agents or if the return on currency speculation is not risky. It should, therefore, always be possible to include a risk premium with the right properties for reconciling the time series – for example, Fama (1984). That is, if currency speculation is risky there is an omitted term in Equation (1) that may be driving the counter-intuitive findings. In this sense, what makes the finding somehow puzzling is not only the sheer magnitude of the deviation from the null but also the incapacity of known asset pricing models to generate the risk premia with the 'right' time series property. A well succeed consumption-based model should have a performance close enough to the results observed using the mimicking portfolio, which represents an upper bound on the pricing capability of any model. So, we take this essence to motivate our pricing kernel construction. Our main innovation regards the procedure which consists in

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[1]Hodrick (1987) and Engel (1996) are very comprehensive surveys.

searching for this mimicking portfolio – a purely statistical pricing kernel that 'solves the puzzle' when one uses Euler equation tests, and then, using it to generate the omitted risk premium term, theoretically responsible for the bias in the log equation.

Although the FPP is characterized both by violations of Euler equation orthogonality tests, that is, in levels, or through the linearized version (Equation (1)), that is, in logs, most studies report it by running the regression above. Relatedly, most studies that address the FPP do so by trying to identify and correct the potential bias on the coefficient of this regression. Our findings in this article indicate that finding a pricing kernel that passes the Euler equation orthogonality tests is not equivalent to 'correcting' Equation (1): the additional assumptions required to linearizing the pricing equation may be too restrictive. Due to the relevance of the UIP as a usual condition in economic models,[2] its failure concerns not only international finance researchers, but all open economy macroeconomists.

The first step in our approach thus consists in devising a preference-free pricing kernel which captures the main sources of risk from a representative set of developed and emerging economies government bonds. We use observable factors which summarize the systematic variation of a global data set mostly composed of (i) aggregate US$ real returns on G7-countries stock indices and short-term government bonds; (ii) returns on 22 (developed and emerging) government bonds; and (iii) returns on Lustig and Verdelhan (2007) 8 foreign dynamic currency portfolios.

The estimated pricing kernel is then used in an in-sample pricing test in level. The time series for the pricing kernel is shown to correctly price the components of the forward premium for each currency (covered and uncovered trading of foreign government bonds). That is, it passes all the Euler equation orthogonality tests as well as the in-sample prediction exercise.

Next, we assume log-normality of discounted returns to derive a system of two equations: one describing the currency depreciation movements and another describing the stochastic discount factor (SDF) movements.[3] The derivation is based on a framework which has the conventional regression used in most empirical studies related to the FPP as a particular case. We characterize and estimate a parsimonious bivariate generalized autoregressive conditional heteroscedasticity-in-mean (GARCH-M) model, following Malmsteen and Teräsvirta (2011) and Goeij and Marquering (2004), where the conditional variance and covariance terms are allowed to affect the mean equation. The estimated system generates a time series for the omitted terms. Our construction allows us to assess whether it is the omission of this term from the usual equation that may be generating a bias in the estimated coefficients.

To summarize our results, we find evidence that, although the omitted term 'explains' the risk premium – that is, the null hypothesis that conditional excess returns are zero is not rejected in Euler equation tests in level – its inclusion in the conventional regression change neither the significance nor the magnitude of the strong predicting power of the forward premium. These results suggest that the assumption of conditional log-normality of discounted returns may be to blame.[4] The disappointing results in the foreign exchange forecasting exercise are suggestive of the potential problems with auxiliary assumptions adopted for linearizing the system.

The remainder of the article is organized as follows. After a brief account of the related literature, Section IV describes and performs the empirical exercises. In Section V, the results are analysed and additional discussion of the role of log-normality is conducted. Section VI concludes.

## II. Related literature

### *FPP: the conventional regression and previous evidences*

Testing the UIP hypothesis using Equation (1) is equivalent to testing the null that $\alpha_1 = 1$ and $\alpha_0 = 0$, along with the uncorrelatedness of residuals from the estimated regression. This is apparent when we write the UIP condition,

$$\mathbb{E}_t(s_{t+1} - s_t) = i_t - i_t^* = f_{t+1} - s_t, \tag{2}$$

[2]The best known is probably Mundell and Mobility (1963).
[3]Log-normality is directly tested and is rejected. Still, the relevant question is whether it is a good enough approximation to allow us to use the regression.
[4]Only for the Canadian dollar, where the conditional covariance term is very volatile, does the inclusion of the omitted terms slightly improve the estimation of the forward premium effect. It does not accommodate the anomaly.

where $i_t$ and $i_t^*$ are, respectively, the nominal net return on a domestic and foreign risk free asset.

Not only is the null rejected in almost all studies, the magnitude of the discrepancy is very large: according to Froot (1990), the average value of $\hat{\alpha}_1$ is −0.88 for over 75 published estimates across various exchange rates and time periods. A negative $\alpha_1$ implies an expected domestic currency appreciation when domestic nominal interest rates exceed foreign interest rates. More importantly, the magnitude of the coefficients suggests a behaviour of a risk premium which is hard to justify with our current models.

This forward premium behaviour has become a well-established regularity. According to Bekaert and Hodrick (1993), these previous inferences are shown to be robust to the measurement error due to incorrect sampling of the data or to failure to account for bid-ask spreads. Korajczyk and Viallet (1992) and Baillie and Bollerslev (2000) are examples of tests that directly use regression (Equation (1)). In the former paper, the authors apply the arbitrage pricing theory (APT) to a large set of assets from many countries and test whether including the factors as the prices of risk reduces the predictive power of the forward premium. They are able to show that the forward premium has lower though not zero predictive power when one includes the factors. Their results should be taken with some caution since they also show that the inclusion of factors often reduces the overall explaining power (as measured by the $R^2$) of the model.

### How the literature tries to account for the FPP and our contribution

Since the UIP need only hold in a world with risk neutrality and/or if returns to speculation in foreign currency are not risky, the absence of a risk adjustment is the 'usual suspect' for the failure of UIP to hold in the data. Accordingly, this is the path that most of the literature has followed.

The use of risk adjustment as an attempt to account for the forward premium anomaly is not new. Using the consumption capital asset pricing model, Mark (1985) estimated extremely high values for the representative agents' risk aversion parameter, in-line with the results found by Hansen and Singleton (1983) and Mehra and Prescott (1985) regarding the equity premium. More recently, Lustig and Verdelhan (2007) used a consumption-based representative agent model which they calibrate by borrowing structural parameters from Yogo (2006) to price eight different foreign assets portfolios built using information on differential interest rates. Although they claim success in accounting for the FPP, their result is disputed by Burnside (2010). We shall not take sides on this issue, for our purposes, it suffices to note that in all these works, asset pricing tests are done in level. That is, they all conduct orthogonality tests for the pricing errors associated with the Euler equations associated with each model.

Because we do not depend on the specific model that generates the time series for the pricing kernel, our only conceptual issue in this first step is whether such kernel exists. All papers mentioned above take this for granted since the perfect markets competitive equilibrium assumption implies that the law of one price. A different view of the puzzle advanced by Burnside (2010) implies otherwise by focusing on market microstructure issues that lead to the violation of the law of one price. Currency markets are thin and characterized by important transaction costs, according to Burnside (2010). He argues that bid-ask spreads are not negligible and that price pressures are also important in understanding the persistence of apparently unexplored opportunities in those markets. Despite acknowledging that microstructure issues may play a role, we shall show that our pricing kernel does a very decent job in pricing returns to currency speculation.

Our methodology, therefore, differs crucially from all these works and from Korajczyk and Viallet (1992) in particular with respect to the choice of the risk adjustment term in the equation. While theirs requires the right choice of risk factors in the APT, ours is based on the explicit (econometric) modelling of the conditional covariance term with a well-adjusted empirical pricing kernel.

The main issue for all these papers is to find risk measures that arise from an underlying economic model. We do not face the same hurdle these works do, because we try not to 'explain' the puzzle. Instead, we devise a statistical kernel as a mimicking portfolio aiming to perform well in in-sample level equation tests, and then we use its estimated conditional covariance time series to try and correct the usual equation.

## III. Foreign government bonds pricing theory

### The SDF and the classic pricing theory in level

Following Michael Harrison and Kreps (1979) and Hansen and Richard (1987), we write the asset pricing equations,

$$1 = \mathbb{E}_t\left[M_{t+1}R^i_{t+1}\right] \quad \forall i = 1, 2, \ldots, N \tag{3}$$

and

$$0 = \mathbb{E}_t\left[M_{t+1}\left(R^i_{t+1} - R^j_{t+1}\right)\right] \quad \forall i, \; j = 1, 2, \ldots, N \tag{4}$$

$\mathbb{E}_t(\cdot)$ denotes conditional expectation given the information available at time $t$. $R^i_{t+1}$ and $R^j_{t+1}$ represent, respectively, the real gross return on assets $i$ and $j$ at time $t$ + 1 and $M_{t+1}$ is the SDF, a random variable that generates unitary prices from asset returns.

Given free portfolio formation, the law of one price is equivalent to the existence of such an SDF through Riesz representation theorem. The correlation with the SDF is the only measure of risk that matters for asset pricing. The law of one price is, therefore, all we need to guarantee the validity of Equations (3) and (4).

Let $M^*_{t+1}$ be the return on the mimicking portfolio, then all $M_{t+1}$ are of the form $M_{t+1} = M^*_{t+1} + \nu_{t+1}$ for some $\nu_{t+1}$ such that $\mathbb{E}_t\left[\nu_{t+1}R^i_{t+1}\right] = 0 \, \forall i$. Therefore, even the assumption that $M^*_{t+1}$ is positive is a restrictive one.[5]

Back to the FPP, one must recall that the covered and uncovered trade of a foreign government bond are themselves portfolios. The asset pricing Equations (3) and (4) applies, therefore, to them. Let the standing representative agent be a US investor who can freely trade domestic and foreign assets.[6] Next, define the covered, $R^C$, and the uncovered return, $R^U$, on foreign government bonds trade,

$$R^C_{t+1} = \frac{{}_tF_{t+1}\left(1 + i^*_t\right)P_t}{S_tP_{t+1}} \quad \text{and} \quad R^U_{t+1} = \frac{S_{t+1}\left(1 + i^*_t\right)P_t}{S_tP_{t+1}}, \tag{5}$$

where ${}_tF_{t+1}$ and $S_t$ are the forward and spot prices of foreign currency in units of domestic currency, respectively, and $P_t$ is the dollar price level. Then, substitute $R^C$ for $R^i$ and $R^U$ for $R^j$ in Equations (3) and (4) to get

$$1 = \mathbb{E}_t\left[M_{t+1}\frac{P_t\left(1 + i^*_t\right){}_tF_{t+1}}{S_tP_{t+1}}\right] \tag{6}$$

and

$$0 = \mathbb{E}_t\left\{M_{t+1}\frac{P_t\left(1 + i^*_t\right)\left[{}_tF_{t+1} - S_{t+1}\right]}{S_tP_{t+1}}\right\}. \tag{7}$$

### The FPP with log-normal discounted returns

If we assume log-normality of discounted uncovered and covered returns, it is straightforward to show that, we can rewrite Equations (6) and (7), obtaining, respectively,

$$m_{t+1} - \pi_{t+1} = -\left({}_tf_{t+1} - s_t + i^*_t\right) - \frac{1}{2}\sigma^2_t(m_{t+1} - \pi_{t+1}) + \nu_{t+1} \tag{8}$$

and

$$s_{t+1} - s_t = \left({}_tf_{t+1} - s_t\right) - \frac{1}{2}\sigma^2_t(s_{t+1} - s_t) - \mathrm{cov}_t(m_{t+1} - \pi_{t+1}, s_{t+1} - s_t) + \varepsilon_{t+1} \tag{9}$$

where $\mathrm{cov}_t(\cdot, \cdot)$ denotes the conditional covariance given the available information at time $t$, $\pi_{t+1}$, the (log of) domestic price variation,[7] $\varepsilon_{t+1}$ denotes the innovation in $\ln\left(M_{t+1}R^U_{t+1}\right) - \ln\left(M_{t+1}R^C_{t+1}\right)$, and $\nu_{t+1}$ denotes the innovation in $\ln\left(M_{t+1}R^C_{t+1}\right)$.

The last terms $\mathrm{cov}_t(m_{t+1} - \pi_{t+1}, s_{t+1} - s_t)$ and $\sigma^2_t(s_{t+1} - s_t)$ can be interpreted as the time-dependent risk premium. Thus, if the correlation between $\left({}_tf_{t+1} - s_t\right)$ and $\mathrm{cov}_t(m_{t+1} - \pi_{t+1}, s_{t+1} - s_t)$ is different from zero the least-square estimates of $\alpha_1$ in Equation (1) are inconsistent. The same happens if correlation between $\left({}_tf_{t+1} - s_t\right)$ and $\sigma^2_t(s_{t+1} - s_t)$ is not zero.

Equation (9) is also useful in deriving empirical tests of FPP. First, one should note that the

[5]Under no arbitrage, we can guarantee that there exists a strictly positive SDF that correctly prices the returns of all assets. Because we are not assuming complete markets, we cannot rely on uniqueness to assert that the return to the mimicking portfolio is non-negative.

[6]Here, we are implicitly assuming the absence of short-sale constraints or other frictions in the economy. Our assumption is in contrast with that of Burnside, Eichenbaum, and Rebelo (2007) for whom bid-ask spreads' impact on the profitability of currency speculation plays the main role in generating the FPP.

[7]Variables in small letters are the logs of the variables in capital letters.

conventional regression (Equation (1)) is a particular case of Equation (9), where the term $1/2\sigma_t^2(s_{t+1}-s_t)+\mathrm{cov}_t(m_{t+1}-\pi_{t+1}, s_{t+1}-s_t)$ is assumed to be time-invariant. In a log-normal world, that is, under Equation (9), it is apparent that the disappointing empirical findings reported when regression (Equation (1)) is used, may be due to a problem of omitted variable. The $\sigma_t^2(s_{t+1}-s_t)$ is the Jensen inequality term, due to our defining the expected rate of depreciation in log. Bekaert and Hodrick (1993) show that the simple omission of this term in Equation (1) cannot be responsible for estimating $\hat{\alpha}_1 < 0$. Therefore, all the action must come from the $\mathrm{cov}_t(m_{t+1}-\pi_{t+1}, s_{t+1}-s_t)$ term. Given that asset returns have clear signs of conditional heteroscedasticity, one is lead to wonder whether this might not be the key to accounting for $\hat{\alpha}_1 < 0$. Note that the covariance term must exhibit considerable variation if it is to explain the evidence regarding the forward premium.[8]

As emphasized in Smith and Wickens (2002), a first important implication that follows from Equations (8) and (9) is that a model satisfying no-arbitrage condition must have a time-varying conditional covariance terms in mean. Second, this approach must be multivariate and the conditional covariance has to be modelled at the same time as the other variables.[9] To incorporate the omitted variable in Equation (1), we use models of the autoregressive conditional heteroscedasticity family, introduced by Engle (1982).[10]

## IV. Our approach

### *Return-based pricing kernel*

As mentioned, our exercise starts with the construction of a time series for an SDF that prices correctly the real returns on uncovered and covered trading with foreign government bonds. Next the system formed by Equations (8) and (9) is estimated using a bivariate GARCH-M approach. This strategy allows us to pin down the conditional covariance and variance terms and enables us to compare with the benchmark results obtained for the traditional ordinary least-squares (OLS) regression, Equation (1).

We must first identify the realized values of a valid pricing kernel. The question is then how to observe these realizations of a pricing kernel? In an influential paper, Hansen and Jagannathan (1991) have led the profession towards return-based kernels. The whole idea is to combine statistical methods with economic theory to devise pricing-kernel estimates that do not depend on preferences but solely on returns. They are projections of SDF in the space of returns, at least to a first-order approximation.

The idea here is to impose a minimum requirement on the behaviour of asset prices. In some sense, one refrains from trying to 'explain' asset prices to focus on guaranteeing that they behave consistently with some agreed upon minimum aspects any theory must satisfy. We shall use this approach which fits our stated purposes perfectly.

Recalling that we have used $M_{t+1}^*$ to represent the return of the mimicking portfolio, the technique employed here to estimate $M_{t+1}^*$ uses principal-component and factor analyses.[11]

Assume that factor models are able to summarize the systematic variation of the $N$ elements of the vector $\mathbf{R}_t = \left(R_t^1, R_t^2, \ldots, R_t^N\right)'$ using a reduced number of $K$ factors, $K < N$. Consider a $K$-factor model in $R_t^i$

$$R_t^i = a_i + \sum_{k=1}^{K} \beta_{i,k} f_{k,t} + \eta_{it} \tag{10}$$

where $f_{k,t}$ are zero-mean pervasive factors and, as is usual in factor analysis, $p\lim\left(1/N\sum_{i=1}^{N}\eta_{i,t}\right) = 0$.

When a multi-factor approach is used, the first step is to specify the factors by means of a statistical or theoretical method and then to consider the

[8]For more details about the heteroscedasticity of asset returns, see Bollerslev, Engle, and Wooldridge (1988), Engle, Ito, and Lin (1990) and Engle and Marcucci (2006).

[9]These requirements rule out many financial econometric techniques, as VAR models. For details, see Campbell, Lo, and MacKinlay (1996).

[10]Alternatively, we could have used Hermite polynomials approaches to provide an approximation for the heteroscedasticity typically found in financial time-series data. Gallant and Tauchen (1989) use the Hermite polynomials as a general approximation to a density function and apply various algebraic simplifications to put it in a tractable form, producing a semi non-parametric approach. Gallant, Hsieh, and Tauchen (1991) follows a similar receipt to model variance as an unobserved stochastic process and then to fit daily observations on exchange rates.

[11]The procedure may be traced back to the work of Ross (1976) and further developed by Chamberlain and Rothschild (1983), Connor and Korajczyk (1986) and Connor and Korajczyk (1993). A recent reference is Bai (2009). The method is asymptotic: either $N \to \infty$ or $N, T \to \infty$, relying on weak law-of-large-numbers to provide consistent estimators of the SDF mimicking portfolio – the unique systematic portion of asset returns.

estimation of the model with known factors. Here, we are not particularly concerned about identifying the factors themselves. Rather, we are interested in constructing an estimate of the SDF, labelled $\tilde{M}_t^*$ by collapsing the factors into a single variable. For that, we shall rely on a purely statistical model, so that, given the equivalence with beta pricing model, we can derive the following linear model for the SDF:

$$\tilde{M}_t^* = a + \sum_{k=1}^{K} b_k f_{k,t} \quad \text{and} \quad \mathbb{E}\left(\tilde{M}_t^* R^i\right) = 1, \quad i = 1, 2, \ldots, N \tag{11}$$

The number of factors used in the empirical analysis is an important issue. We expect $K$ to be rather small, but have some flexibility for this choice. We took a pragmatic view here, increasing $K$ until the estimate of $M_t^*$ changed very little due to the last increment; see for instance Lehmann and Modest (1988), Connor and Korajczyk (1988) and Tsay (2001).

Once we have recovered our multi-factor $\tilde{M}_t^*$ and shown that its time series approximates well enough that of the pricing kernel, we can use it to perform the pricing test in level. These tests should be viewed as an in-sample exercise, since $M^*$ was estimated with representative global assets alone and then used to price the foreign assets. Once we are confident that our pricing kernel does a good job in pricing the relevant foreign assets in-sample, we then proceed to the estimation of the GARCH-M approach, modelling exchange rate movements under the log-normality assumption.

## *Pricing test in level*

Our pricing tests in level rely on two variants of the pricing equation, given our intent to model covered and uncovered trading with foreign government bonds:

$$0 = \mathbb{E}\left\{ \begin{bmatrix} \tilde{M}_t^* \frac{P_t\left(1+i_{t+1}^*\right)\left[{}_tF_{t+1} - S_{t+1}\right]}{S_t P_{t+1}} - \mu_1 \\ \tilde{M}_t^* \frac{S_{t+1}\left(1+i_{t+1}^*\right)P_t}{S_t P_{t+1}} - \left(1 - \mu_2\right) \end{bmatrix} \times \boldsymbol{z_t} \right\} \tag{12}$$

where we must consider $\boldsymbol{z_t}$ to be a vector of instrumental variables, which are all observed up to time $t$, therefore measurable with respect to $\mathbb{E}_t(\cdot)$.

The system (Equation (12)) of orthogonality restrictions can be used to assess the pricing behaviour of estimates of $M^*$ with respect to the components of the forward premium or any linear rotation of them. Equations in the system can be tested separately or jointly. In testing, we employ a generalized method-of-moment (GMM) perspective, using this system as a natural moment restriction to be obeyed and assuming that parameters $\mu_1$ and $\mu_2$ should be interpreted as pricing errors. We assume that there are enough elements in the vector $\boldsymbol{z_t}$ for $\mu_1$ and $\mu_2$ for the system to be over-identified. In order for Equations (6) and (7) to hold, we must have $\mu_1 = 0$ and $\mu_2 = 0$, and the over-identifying restriction $T \times J$ test in Hansen (1982) should not reject them. The latter can be viewed as testing whether or not scaled pricing errors are statistically zero.

Our estimates in the main asset pricing test reported here in Table 1 are from a GMM, not with an efficient GMM (iterated to convergence), but produced by an iterate procedure. We chose it since it usually performs better in practice, resulting in asymptotically equivalent estimates with better small-sample

**Table 1.** FPP test: in-sample asset pricing exercise for the foreign currencies.[a,b,c,d]

| FPP test using $\tilde{M}_{t+1}^*$: $\mathbb{E}_t\left\{\begin{array}{c}\tilde{M}_{t+1}^* \cdot \left(R_{t+1}^{U,j} - R_{t+1}^{C,j}\right) - \mu_1 \\ \tilde{M}_{t+1}^* \cdot R_{t+1}^{U,j} - (1-\mu_2)\end{array}\right\} = 0$ | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Instrument set: $\left({}_tF_{t+1}^j - S_t^j\right)/S_t^j$ and a constant | | | | | | | | |
| Wald test (deviations) | | | Asset pricing and deviations | | | | Overall fit | |
| Results for British government bonds: | | | | | | | | |
| $H_0$: $\mu_i = 0,\ i = 1,2$ | | | $\hat{\mu}_1$ | −0.012 | (0.611) | [0.984] | *J*-statist. | 0.011 |
| $\chi^2$ | 0.015 | [0.993] | $\hat{\mu}_2$ | −0.452 | (3.771) | [0.905] | *p*-value | [0.540] |
| Results for Canadian government bonds: | | | | | | | | |
| $H_0$: $\mu_i = 0,\ i = 1,2$ | | | $\hat{\mu}_1$ | −0.187 | (0.242) | [0.440] | *J*-statist. | 0.044 |
| $\chi^2$ | 0.660 | [0.719] | $\hat{\mu}_2$ | −1.871 | (3.989) | [0.639] | *p*-value | [0.085] |
| Results for German government bonds: | | | | | | | | |
| $H_0$: $\mu_i = 0,\ i = 1,2$ | | | $\hat{\mu}_1$ | −0.437 | (0.631) | [0.489] | *J*-statist. | 0.061*** |
| $\chi^2$ | 1.744 | [0.418] | $\hat{\mu}_2$ | 2.782 | (3.627) | [0.444] | *p*-value | [0.033] |
| Results for Japanese government bonds: | | | | | | | | |
| $H_0$: $\mu_i = 0,\ i = 1,2$ | | | $\hat{\mu}_1$ | 0.485 | (0.664) | [0.466] | *J*-statist. | 0.108*** |
| $\chi^2$ | 0.683 | [0.711] | $\hat{\mu}_2$ | 1.604 | (3.859) | [0.678] | *p*-value | [0.002] |
| Results for Swiss government bonds: | | | | | | | | |
| $H_0$: $\mu_i = 0,\ i = 1,2$ | | | $\hat{\mu}_1$ | −1.064 | (0.801) | [0.185] | *J*-statist. | 0.029 |
| $\chi^2$ | 1.772 | [0.412] | $\hat{\mu}_2$ | −0.867 | (3.779) | [0.819] | *p*-value | [0.197] |

***Indicates the rejection of the validity of the over-identifying restrictions ($T \times J$ statistic) at 5% significance level.

[a]Hansen's (1982) GMM technique (estimates produced by an iterate procedure) used to test Euler equations and to estimate the model parameters, over the period from 1977:1 to 2004:4, 112 observations.

[b]We use the superscript *j* for returns, forward and spot rates in order to associate these variables to the country *j*.

[c]Respective standard errors are reported in the parenthesis while *p*-values in the box brackets.

[d]The pricing errors and the standard errors are reported multiplied by 100.

properties according to Ferson and Foerster (1994). For a full discussion, see Cochrane (2001). In order to deal with the possible difficulty to achieve an optimal weighting scheme in GMM estimation in this article, since we have 2 pricing equations with 2 instruments (4 orthogonality conditions) estimated using 112 time-series observations, we also use the identity matrix in weighting orthogonality conditions as a robustness check, correcting the estimate of their asymptotic variance–covariance matrix for the presence of serial correlation and heteroscedasticity of unknown form using the techniques in Newey and West (1987).

### Conditional heteroscedasticity models

Consider Equations (8) and (9). Replacing the pricing kernel by its estimate, we have the following system:

$$\begin{cases} s_{t+1} - s_t = ({}_tf_{t+1} - s_t) - \frac{1}{2}\sigma_t^2(s_{t+1} - s_t) - \mathrm{cov}_t(\tilde{m}^*_{t+1} - \pi_{t+1}, s_{t+1} - s_t) + \varepsilon_{t+1} \\ \tilde{m}^*_{t+1} - \pi_{t+1} = -({}_tf_{t+1} - s_t + i^*_t) - \frac{1}{2}\sigma_t^2(\tilde{m}^*_{t+1} - \pi_{t+1}) + \nu_{t+1}, \end{cases} \tag{13}$$

where both process, $s_{t+1} - s_t$ and $\tilde{m}^*_{t+1} - \pi_{t+1}$ depend on some of the conditional variance–covariance matrix terms. This characterizes a bivariate GARCH-in-mean model. This specification, proposed by Engle, Lilien, and Robins (1987) although not based in economic theory, provides a good approximation to the heteroscedasticity typically found in financial time-series data. It extends the family of autoregressive conditional heteroscedasticity models to allow the conditional variance to affect the mean.[12]

Taking parsimony, simplicity and our small-sample in the time-series dimension into account we use bivariate **GARCH-M(1,1)** to obtain a better adjusted conditional volatility model for each process.[13]

Another issue we have to face is the possible non-positiveness of the estimated covariance matrix, as emphasized in this following quote from Goeij and Marquering (2004), '[…] In the univariate GARCH (1, 1) the restrictions on the parameter to ensure that the variance is non-negative are quite obvious. However, to guarantee that the conditional covariance matrix is positive definite is less straightforward in a multivariate setting. […] If this happens [the covariance matrix become non-positive during the estimation procedure], the estimated covariance matrix cannot be inverted, and the maximum-likelihood method fails to compute an optimum'.

The specification used here to circumvent this non-positivity problem follows the **BEKK** model,[14] developed by Baba, Engel, Kraft, and Krone (1993). Its improvement upon other models of this class is that the representation for the matrix $H_t$ guarantees it is positive definite for all values of $t$ and, additionally, there are less parameters to be estimated when compared to traditional VECH representation. Another advantage of this representation is that it does not impose any restrictions on the form of the conditional covariances. (For instance, the method proposed by Bollerslev (1990) which imposes constant conditional correlation (CCC-GARCH model).)

We compare the OLS and GARCH-M estimates. The omitted variables, $\sigma_t^2(s_{t+1} - s_t)$ and $\mathrm{cov}_t(m_{t+1} - \pi_{t+1}, s_{t+1} - s_t)$, are responsible for the bias on the estimation of Equation (1) if the GARCH-M estimate of forward premium parameter is closer to unit than the OLS estimate.

## V. Empirical exercise

### Data and summary statistics

#### Foreign government bonds trading

We collect foreign currency data in American dollars for Canada, Germany, Japan, Switzerland and the United Kingdom, covering the period starting in 1977:1 and ending in 2004:4; a total 112 observations per country with quarterly frequency. Forward rate series were extracted from the Chicago Mercantile of Exchange database, and spot rate series from the Bank of England database. We compute the US$ real returns on short-term British, Canadian, German, Swiss and Japanese government bonds, using both spot and forward exchange-rate to

[12]According to Dominguez (1998), most studies that have focused on the effects of intervention on exchange rate volatility, looked at conditional exchange rate volatility, usually estimated with GARCH framework.

[13]See, for example, West and Cho (1995) and Malmsteen and Teräsvirta (2011).

[14]A drawback of the BEKK model for our purposes is that its extension towards asymmetric versions is not straightforward.

transform returns denominated in foreign currency into US$.

Observing summary statistics of our database over the period 1977:1 to 2004:4, the real return on covered trading of foreign bonds show means and standard deviations quite similar to that of the US Treasury Bill (T-Bill), which is reasonable, since it reflects the covered interest rate parity in a frictionless economy. Regarding the return on covered trading, the means range from 0.31% to 0.66%, and standard deviation, from 0.86% to 1.10%, except for the Swiss franc where these values are 2.06% and 7.74%, respectively.

The forward rates for the Swiss franc in Chicago Mercantile Exchange database when compared with the respective spot rates, display an outlier in 1995. This forward series is not, however, used to estimate the pricing kernel and the outlier issue turned out not to be relevant in our exercise. Indeed, when we use the outlier-removed sample to check the robustness of our empirical results, they turned out to be qualitatively identical for these two different sets of data.

Regarding the return on uncovered trading, averages range from 0.68% to 1.26%, while standard deviations range from 2.77% to 8.10%. Downside risk, which takes into account only negative deviations, range from 0.58% to 1.50% and from 1.79% to 5.29%, considering covered and uncovered trading, respectively. As for the location and the variability of the returns, except for the covered trading with British and Canadian bonds, we can observe that the data are skewed right meaning that the right tail is longer than the left one and for most cases, the high values for kurtosis indicate real returns are peaked relative to a normal distribution.

Finally, we also observe modest values for Sharpe and Sortino ratios (using T-Bill as risk-free asset), relative to these measures for S&P500, or other stock market benchmarks.

### Pricing kernel

To 'construct' the return-based estimate of the pricing kernel – $\tilde{M}^*_{t+1}$ – we use a *global* data set composed of aggregate US$ real returns on G7-country stock indices, on short-term government bonds and on 22 (developed and emerging) government bonds. Spot exchange rate data were used to transform returns denominated in foreign currency into US$ and the consumer price index of services and non-durable goods in the United States was used as a deflator. In addition to this data set, we also use US$ real returns on Lustig and Verdelhan (2007) eight foreign dynamic currency portfolios. These series were built using quarterly series of spot exchange rates and short-term foreign government bonds available in IFS database.

Following Lustig and Verdelhan (2007), the interest rate on a 3-month government security (e.g. a US T-Bill) or an equivalent instrument is used as the foreign interest rate. As data became available, new countries are added (or subtracted) from these portfolios as we build eight different zero-cost foreign-currency portfolios, by selling 90 day T-Bills and using the proceeds to buy government bonds of various countries. Portfolios are ranked according to interest rate differential with respect to the T-Bill – that is, excess returns here in units of foreign currency over units of US currency. As reported in da Costa, Issler, and Matos (2015), some of these foreign-currency portfolios have a negative real excess return but, in general, their volatilities seem to be small.

These portfolios cover a wide spectrum of investment opportunities across the globe. This is an important element of our choice of assets, since diversification is recommended for both methods of computing $M^*_t$. Despite the reduced number of *global* portfolios to estimate $M^*_t$, it is plausible to expect the operation of a weak-law of large numbers, since these portfolios contain hundreds of representative assets. Therefore, the final average containing these portfolios will be formed using thousands of assets worldwide. Of course, it would be preferable to work at a more disaggregated level. However, we know no comprehensive database with a long enough time span containing worldwide returns at the firm level, for instance.

It is important to bear in mind that the exclusion of potential assets from our sample does not mean that the agents did not trade on these assets. This is something often forgotten in studies involving the financial markets but the point is that the 'parameter' $M^*_t$ we want to estimate is not altered by our restricting the information we use in estimating it! The problem is not, therefore, what the space

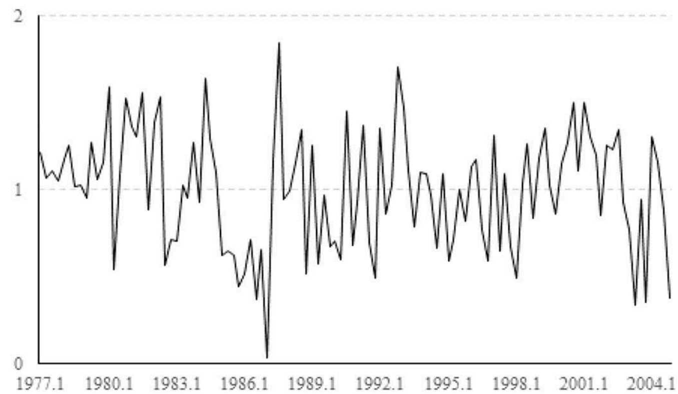


**Figure 1.** Multi-factor pricing kernel. $\tilde{M}^*_{t+1}$ is constructed from the US$ real returns on some of the most relevant global assets: G7-country stock indices and short-term government bonds, 22 (developed and emerging) government bonds and on Lustig and Verdelhan (2007) 8 foreign dynamic currency portfolios, constructed by da Costa, Issler, and Matos (2015).
Source: Data from 1977:1 to 2004:4, 112 observations.

spanned by each set of asset is, as one might have been lead to think, but whether the observed assets are representative in the sense of satisfying the assumptions presented in the previous subsections: Are we capable of identifying the realized SDF from the observation of a sample of assets?

In extracting a time series for $\tilde{M}^*_{t+1}$, we need to choose the number of factors. Following most of the empirical literature, we took a pragmatic, informal, but useful view, examining the time plot of the eigenvalues ordered from the largest to the smallest. We found that 6 factors were needed to account for a representative portion of the variation (81.2%) of our 44 dynamic portfolio returns. The estimated $\tilde{M}^*_{t+1}$ is plotted in Figure 1.

It is apparent that the multi-factor model generates an SDF that (i) is more volatile (0.34) than the one obtained from canonical consumption-based models and (ii) has a mean that is in-line with the literature, since the respective unconditional return on risk free asset would reach slightly negative values, even when annualized.

## *Pricing test*

Before the actual pricing tests, we conducted tests on returns and its discounted series for the presence of a unit root in the autoregressive polynomial.[15] Because returns are prone to displaying heteroscedasticity, we used the Phillips and Perron (1988) unit-root test including an intercept in the test regression. According to the results for this test, for all cases we rejected the null of a unit root at the 1% level. We take this as evidence that all returns and the pricing kernels are well behaved.[16]

To try to isolate the hole played by the log-normality hypothesis, we first verify the behaviour of our return-based SDF, $\tilde{M}^*_{t+1}$, in pricing covered and uncovered trading with the foreign government bonds using Euler equation orthogonality tests. This is an in-sample pricing test, which can be also seen as a law-of-one-price test for the foreign market: a test performed when a global SDF $\tilde{M}^*_t$ is used in pricing foreign government bonds. Table 1 presents the results of our 'in level' exercise.

Our FPP 'in level' approach tests the system of orthogonality restrictions (Equation (12)).

There are two Euler equations: one with the excess return on the uncovered over the covered trade with foreign government bonds $\left(R^{U,j}_{t+1} - R^{C,j}_{t+1}\right)$ and one with the uncovered return $R^{U,j}_{t+1}$. The instrument set includes a constant and the variable that generates the FPP in regression (Equation (1)), that is, the respective $\left({}_tF^j_{t+1} - S^j_t\right)\Big/S^j_t$, making the result of these tests very informative regarding the FPP. Pricing errors and the respective standard errors are multiplied by 100.

[15]See Hansen (1982) and Ogaki (1992) for more details.
[16]These results are robust to the Augmented Dickey–Fuller technique, also including intercept and/or trend.

For all foreign currencies studied here, the results of these pricing tests are quite similar: at the 5% significance level, all individual pricing errors are statistically zero and the Wald test results show no sign of joint significance for currency pricing errors. Identical results are observed for all the over-identifying restriction tests, except for the German and Japanese cases for which the $T \times J$ statistics reject the null that weighted pricing errors are insignificant. These results are robust to the inclusion of additional instruments, including lags of returns in question or macroeconomic variables.

Our results are not due to increased variance. Indeed, when using the multi-factor SDF estimate, the discounted return, that is, scaled by $M^{*}$, displayed *lower* variance than the undiscounted return. The size of the pricing errors is relatively modest. In absolute value, except for the German, Canadian and Japanese uncovered returns, all of these errors take on values lower than one, with a mean value of 0.5, when trying to price returns and excess returns simultaneously. Thus, the high $p$-values found in this article come from lower pricing errors, *not* higher variance. The GMM standard errors in Table 1 range from 0.24% to 0.80% for excess returns and assume values lower than 4.0% for uncovered trading, while the respective values range from 2.50% to 9.78% and from 2.77% to 8.10%.

We also perform robustness checks on the GMM specification. The estimates in the main asset pricing approach do not change much when an identity matrix in weighting moment restrictions GMM specification is used, instead of an iterate procedure.

Our claim that our pricing kernel is able to account for foreign government bond returns in an Euler (in level) framework is strengthened when one observes the pricing error measure based on $\chi^2$ statistics developed by Hansen and Jagannathan (1997), 0.34, and its prediction performance.

In this sense, Figure 2 displays predicted versus realized average return (to be precise, net return) on covered and uncovered trading for our pricing kernel. Pricing errors are measured by the distance to the 45-degree line (dotted line).

Except for the huge prediction error observed for the covered operation with Swiss bonds, it is clear that the pricing kernel does a good job in predicting other assets, with errors lower than 0.4% per quarter. As an additional evidence is useful to convince that our measured pricing kernel is appropriate for subsequent analysis, it does a good job when we use it to account for other international markets stylized facts that escape consumption-based models. For

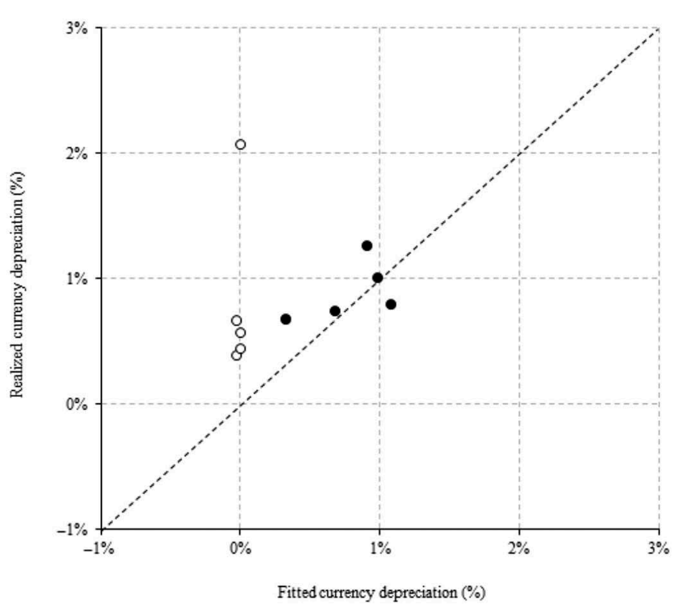


**Figure 2.** $\tilde{M}_t^*$ in-sample prediction: foreign government bond markets. This figure plots the quarterly average return on uncovered (filled dots) and covered (empty dots) trading of British, Canadian, German, Japanese and Swiss government bonds versus its respective predictions of unconditional multi-factor pricing kernel.

Source: Data from 1977:1 to 2004:4, 112 observations.

instance, we can evidence that our SDF prices correctly all the Lustig and Verdelhan (2007) portfolios.[17]

### The log-normality assumption

We first check whether the assumption of log-normality of the underlying processes holds. According to the results based on Lilliefors, Cramer–von Mises, Watson and Anderson–Darling techniques, the discounted covered and uncovered returns do not seem to follow an unconditional log-normal distribution at 1% significance level. We also observe the results for best fitting probability distribution functions and the histograms for returns on uncovered and covered trading. According to the log-likelihood statistic, the best fitting distribution is a 'normal' for all discounted returns in question and the second best distribution is the Weibull for most cases or the smallest extreme values. The log-normal distribution is never characterized as the best one, even if one use the Chi-square or the Kolmogorov–Smirnov statistics, instead of the log likelihood. Observing the skewness measures for the discounted returns in log, lower than −3.0 for all cases, it is clear that they are skewed left, the opposite of what we would expected from data generated from a log normal distributions.

Returns of assets in our data set display the same type of departure from log-normality reported in the literature. The relevant question is whether this departure from log-normality is the underlying reason for the empirical failure of most attempts to fix Equation (1).

### Regressions results: OLS and GARCH-M

In Table 2, we report the estimates of the conventional regression (OLS). Observing the first exercise results, only for Japanese case the hypothesis $\hat{\alpha}_0 = 0$ can be rejected. Regarding the $\hat{\alpha}_1$, one can see that pointwise estimates takes negative values for three of the five currencies. For Japan and Switzerland, one cannot reject its negative sign significance at the 5% level. Only for Canada, we are not able to reject $\hat{\alpha}_1 = 1$. These results are in accordance with the previous results in the literature.

The results for the augmented regression (GARCH-M) are reported in Table 3.

According to the results reported in Table 1, the pricing kernel estimated produces very satisfactory

**Table 2.** FPP test: the conventional regression estimation.[a,b]

Conventional regression: $s_{t+1} - s_t = \alpha_0 + \alpha_1({}_tf_{t+1} - s_t) + u_{t+1}$

| Wald tests | | | | | | Parameters | | | | Other results |
|---|---|---|---|---|---|---|---|---|---|---|
| Results for British government bonds: | | | | | | | | | | |
| $H_0$: $\alpha_1 = 1$ | | | $H_0$: $\alpha_0 = 0,\ \alpha_1 = 1$ | | | $\hat{\alpha}_0$ | −0.002 | (0.007) | [0.765] | $F$-stat.: 1.987 [0.087] |
| $\chi^2$ | 2.269 | [0.135] | $\chi^2$ | 1.653 | [0.196] | $\hat{\alpha}_1$ | −0.594 | (1.0578) | [0.576] | $R^2$ (adj.): 0.004 |
| Results for Canadian government bonds: | | | | | | | | | | |
| $H_0$: $\alpha_1 = 1$ | | | $H_0$: $\alpha_0 = 0,\ \alpha_1 = 1$ | | | $\hat{\alpha}_0$ | 0.000 | (0.003) | [0.888] | $F$-stat.: 0.984 [0.431] |
| $\chi^2$ | 1.527 | [0.219] | $\chi^2$ | 0.782 | [0.460] | $\hat{\alpha}_1$ | 0.580 | (0.340) | [0.091] | $R^2$ (adj.): 0.022 |
| Results for German government bonds: | | | | | | | | | | |
| $H_0$: $\alpha_1 = 1$ | | | $H_0$: $\alpha_0 = 0,\ \alpha_1 = 1$ | | | $\hat{\alpha}_0$ | 0.008 | (0.007) | [0.209] | $F$-stat.: 1.883 [0.103] |
| $\chi^2$ | 12.906** | [0.001] | $\chi^2$ | 6.455** | [0.002] | $\hat{\alpha}_1$ | −0.842 | (0.513) | [0.103] | $R^2$ (adj.): 0.009 |
| Results for Japanese government bonds: | | | | | | | | | | |
| $H_0$: $\alpha_1 = 1$ | | | $H_0$: $\alpha_0 = 0,\ \alpha_1 = 1$ | | | $\hat{\alpha}_0$ | 0.025* | (0.006) | [0.000] | $F$-stat.: 1.883 [0.103] |
| $\chi^2$ | 24.466** | [0.000] | $\chi^2$ | 13.786** | [0.000] | $\hat{\alpha}_1$ | −2.072* | (0.621) | [0.001] | $R^2$ (adj.): 0.009 |
| Results for Swiss government bonds: | | | | | | | | | | |
| $H_0$: $\alpha_1 = 1$ | | | $H_0$: $\alpha_0 = 0,\ \alpha_1 = 1$ | | | $\hat{\alpha}_0$ | 0.003 | (0.008) | [0.735] | $F$-stat.: 0.592 [0.706] |
| $\chi^2$ | 23.480** | [0.000] | $\chi^2$ | 12.248** | [0.000] | $\hat{\alpha}_1$ | 0.236 | (1.578) | [0.136] | $R^2$ (adj.): 0.027 |

*Indicates the rejection of the null hypothesis of insignificant parameter at 5% level.
**Indicates the rejection of null hypothesis of Wald test at 5% significance level.
[a]OLS technique used to estimate the conventional regression parameters over the period from 1977:1 to 2004:4, taking into account the covariance estimate proposed by Newey and West (1987).
[b]Respective standard errors are reported in the parenthesis while $p$-values in the box brackets.

[17]In this specific case, in order to deal with the difficulty to achieve an optimal weighting scheme in GMM estimation in this case (16 orthogonality conditions), we use the identity matrix in weighting orthogonality conditions. For the same reason, additional tests, as pricing all currencies jointly, cannot be performed.

**Table 3.** FPP test: the asset pricing theory-based regression estimation.[a,b,c]

GARCH-M system: $\begin{cases} s_{t+1}-s_t=\alpha_1\left({}_tf_{t+1}-s_t\right)+\beta_{11}\frac{\sigma_t^2(s_{t+1}-s_t)}{2}+\beta_{12}\mathrm{cov}_t\left(\tilde{m}^*_{t+1}-\pi_{t+1},s_{t+1}-s_t\right)+\varepsilon_{t+1} \\ \tilde{m}^*_{t+1}-\pi_{t+1}=\alpha_2\left({}_tf_{t+1}-s_t+i^*_t\right)+\beta_{22}\frac{\sigma_t^2\left(\tilde{m}^*_{t+1}-\pi_{t+1}\right)}{2}+v_{t+1}, \end{cases}$

| Wald tests | | | Parameters | | | |
|---|---|---|---|---|---|---|
| Results for British government bonds: | | | | | | |
| $H_0$: $\alpha_1=1$ — $\chi^2$ 16.459** [0.000] | $H_0$: $\beta_{11}=-\frac{1}{2}$, $\alpha_1=1$, $\beta_{12}=-1$ — $\chi^2$ 25.846** [0.000] | $H_0$: $\alpha_2=-1$, $\beta_{22}=-\frac{1}{2}$ — $\chi^2$ 46.373** [0.000] | $\hat{\alpha}_1$ | −0.857 | (0.458) | [0.061] |
| | | | $\hat{\beta}_{11}$ | −7.996* | (3.086) | [0.009] |
| | | | $\hat{\beta}_{12}$ | −0.946* | (0.453) | [0.037] |
| | | | $\hat{\alpha}_2$ | 1.029 | (1.609) | [0.522] |
| | | | $\hat{\beta}_{22}$ | 0.012 | (0.087) | [0.893] |
| Results for Canadian government bonds: | | | | | | |
| $H_0$: $\alpha_1=1$ — $\chi^2$ 3.273** [0.070] | $H_0$: $\beta_{11}=-\frac{1}{2}$, $\alpha_1=1$, $\beta_{12}=-1$ — $\chi^2$ 80.101** [0.000] | $H_0$: $\alpha_2=-1$, $\beta_{22}=-\frac{1}{2}$ — $\chi^2$ 47.127** [0.000] | $\hat{\alpha}_1$ | 0.514* | (0.268) | [0.050] |
| | | | $\hat{\beta}_{11}$ | −5.600 | (4.474) | [0.211] |
| | | | $\hat{\beta}_{12}$ | −0.076 | (0.253) | [0.762] |
| | | | $\hat{\alpha}_2$ | 0.580 | (1.514) | [0.701] |
| | | | $\hat{\beta}_{22}$ | 0.068 | (0.094) | [0.477] |
| Results for German government bonds: | | | | | | |
| $H_0$: $\alpha_1=1$ — $\chi^2$ 9.826** [0.002] | $H_0$: $\beta_{11}=-\frac{1}{2}$, $\alpha_1=1$, $\beta_{12}=-1$ — $\chi^2$ 33.504** [0.000] | $H_0$: $\alpha_2=-1$, $\beta_{22}=-\frac{1}{2}$ — $\chi^2$ 32.577** [0.000] | $\hat{\alpha}_1$ | −0.676 | (0.535) | [0.206] |
| | | | $\hat{\beta}_{11}$ | −0.683 | (2.336) | [0.770] |
| | | | $\hat{\beta}_{12}$ | −0.115 | (0.349) | [0.742] |
| | | | $\hat{\alpha}_2$ | 2.021 | (1.377) | [0.142] |
| | | | $\hat{\beta}_{22}$ | −0.007 | (0.142) | [0.960] |
| Results for Japanese government bonds: | | | | | | |
| $H_0$: $\alpha_1=1$ — $\chi^2$ 8.566** [0.003] | $H_0$: $\beta_{11}=-\frac{1}{2}$, $\alpha_1=1$, $\beta_{12}=-1$ — $\chi^2$ 15.131** [0.002] | $H_0$: $\alpha_2=-1$, $\beta_{22}=-\frac{1}{2}$ — $\chi^2$ 11.163** [0.004] | $\hat{\alpha}_1$ | −0.937 | (0.662) | [0.157] |
| | | | $\hat{\beta}_{11}$ | 2.238 | (1.694) | [0.186] |
| | | | $\hat{\beta}_{12}$ | −0.341 | (0.474) | [0.472] |
| | | | $\hat{\alpha}_2$ | 2.201 | (3.087) | [0.476] |
| | | | $\hat{\beta}_{22}$ | −0.259 | (0.229) | [0.255] |
| Results for Swiss government bonds: | | | | | | |
| $H_0$: $\alpha_1=1$ — $\chi^2$ 11.256** [0.001] | $H_0$: $\beta_{11}=-\frac{1}{2}$, $\alpha_1=1$, $\beta_{12}=-1$ — $\chi^2$ 21.283** [0.000] | $H_0$: $\alpha_2=-1$, $\beta_{22}=-\frac{1}{2}$ — $\chi^2$ 10.045** [0.006] | $\hat{\alpha}_1$ | 0.457* | (0.162) | [0.005] |
| | | | $\hat{\beta}_{11}$ | −1.385 | (1.468) | [0.345] |
| | | | $\hat{\beta}_{12}$ | −0.169 | (0.488) | [0.729] |
| | | | $\hat{\alpha}_2$ | 0.448 | (1.008) | [0.656] |
| | | | $\hat{\beta}_{22}$ | −0.204 | (0.250) | [0.415] |

*Indicates the rejection of the null hypothesis of insignificant parameter at 5% level.

[a]Bivariate GARCH-M (1,1) – BEKK specification – used to estimate the asset pricing theory-based regression parameters over the period from 1977:1 to 2004:4.

[b]Respective standard errors are reported in the parenthesis while *p*-values in the box brackets.

results for the in-sample pricing test in level, but the same cannot be said for the GARCH-M approach. The omitted terms, which theoretically explain the risk premium, do not seem to be able to account for the deviation from the null.

For most cases, the results obtained for the currency depreciation equation are very different from the values under the null hypotheses $\left(\tilde{\alpha}_1=1,\ \tilde{\beta}_{11}=-1/2 \text{ and } \tilde{\beta}_{12}=-1\right)$, and are rejected for all but the Canadian government bonds, where the coefficient related to the forward premium, $\tilde{\alpha}_1$, is positive and statistically close to one. All rejections at the 5% significance level. Rejections are also observed for the pricing kernel equation, where for the five economies, $\tilde{\alpha}_2\neq-1$ and $\tilde{\beta}_{22}\neq-1/2$.

In a log-normal world, the role played by $\mathrm{cov}_t(s_{t+1}-s_t,m_{t+1}-\pi_{t+1})$ is key. To a minor extent, the conditional variance, $\sigma_t^2(s_{t+1}-s_t)$ may also matter. The whole idea of using a GARCH-M approach is to be able to estimate these terms that are then included in the extended version of the usual equation, for these are the terms that are, in theory at least, responsible for the potential bias on the estimation of $\alpha_1$.

The conditional covariance does not appear to be smooth enough to be omitted, or held constant: the assumption underlying the conventional regression. It fluctuates throughout the whole period from 0.00 to −0.03, except for isolated extreme values. This is of a considerable order of magnitude when currency depreciation, which ranges from −0.05 to 0.05 for most cases, is considered. The conditional variance related to currency depreciation, an inequality Jensen term, is smoother, ranging often from 0.00 to 0.01, but not irrelevant.

Except for the higher values observed for the Swiss and British cases, correlations have absolute values that are lower than 0.06. The detailed behaviour of the conditional variance and covariance time series are not shown here, but they are available upon request.

## The role of log-normality in prediction exercise

The omitted term accounts for the risk premium, based on the asset pricing theory in level. However, the augmented regression proposed here does not have a satisfactory performance in explain the strong negativity of the forward premium effect.

The additional underlying assumption in our procedure is that a parsimonious specification of GARCH-M can correctly capture the dynamics in question, an almost consensual opinion in the empirical finance literature, as we can see in West and Cho (1995) and Malmsteen and Teräsvirta (2011). This assumption, proposed by Engle, Lilien,

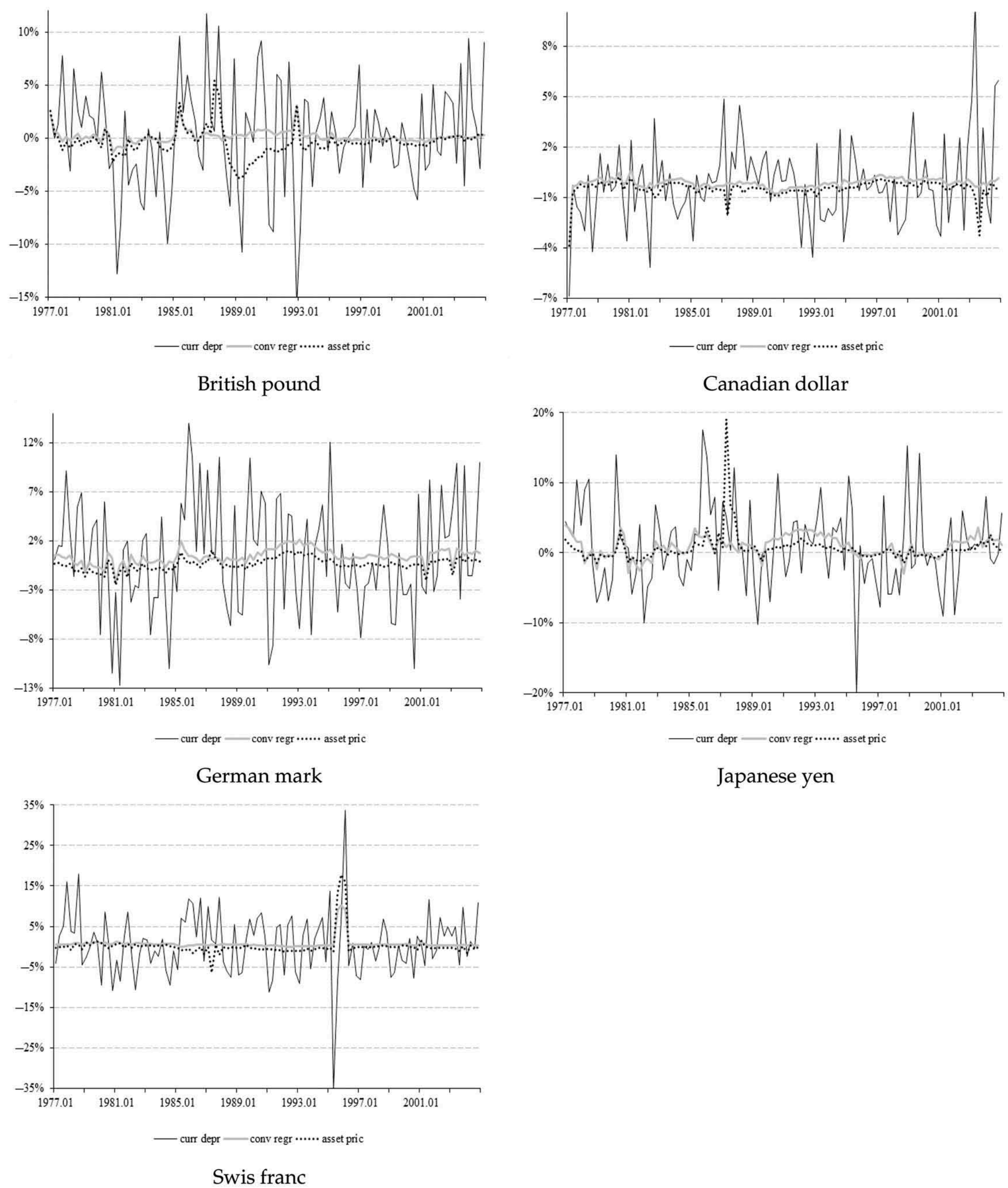


**Figure 3.** The (log of) currency depreciation movement and its in-sample prediction. This figure plots the quarterly currency depreciation movement for the British pound, Canadian dollar, German mark, Japanese yen and Swiss franc versus its respective predictions based on conventional and asset pricing-based approaches.

Source: Data from 1977:1 to 2004:4, 112 observations.

and Robins (1987) and widely used in empirical finance literature, is necessary to be convinced that our exercise warrants our conclusion.

This undesirable role played by the conditional log-normality of discounted returns remains in foreign exchange rates forecasting exercise. According to the evidence displayed in Figure 3, neither the conventional nor the augmented regression seems to be unable to predict quarterly currency depreciation for developed economies in the last decades. Comparing these two methodologies, except for the Japanese yen, the former test performs still better than the GARCH-M-based one, but both with the prediction errors in the same magnitude order when compared with a random walk strategy, $\mathbb{E}_t(s_{t+1}) = s_t$, ranging from 1.8% to 8.5%. Our attempt using GARCH-M approach has been relatively less successful in predicting foreign currency movements compared to Lee (1988) and Baillie and Bollerslev (1990).

## VI. Conclusion

When we depart from the assumption of risk neutrality, the regression typically used to test the whether UIP holds is plagued by biases related to an omitted variable.

In this article, we tried to fix this problem by deriving, under the assumption of log-normality of returns the omitted term in the equation. Our procedure was to construct a pricing kernel that passes all the orthogonality tests used to evaluate its validity in levels. Next, we expressed the omitted variable in the usual equation as a covariance term involving this pricing kernel. We modelled its time series using a GARCH-M approach and ran the regression with this term.

Our results proved disappointing. For most currencies, the term did not correct the bias in the equation. There are many ways one can think about improving our procedure. The most obvious is to consider weaker assumption on the time series of returns.

These approaches to test this equation which is the linearized version of the UIP relation involve considerably higher complexity and do not exempt one from making additional assumptions beyond what is needed to test the orthogonality restrictions that characterize tests in levels. One is then left to wonder whether this is the right path to follow or whether it is not time to abandon the usual equation.

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